

Village Education Resource Center (VERC)

Technical & Operational System Architecture, Data Flows, and Cloud Infrastructure Specifications

SYSTEM VERSION

v2.4.0 (Enterprise Next.js + NestJS)

ARCHITECTURE STYLE

Decoupled Modular Monolith + SSR

DATABASE TIER

PostgreSQL 16 / SQLite Prisma ORM

AI ENGINE

Gemini 1.5 Flash (SSE Streaming)

12+ Modules

ADMIN CAPABILITIES

Real-time CRUD & sync

< 120 ms

AI STREAM LATENCY

Gemini Flash SSE engine

99.10%

MF RECOVERY RATE

112k+ Active Members

Zero-Downtime

CI/CD DEPLOYMENTS

Automated Render & Netlify

1. Executive Summary & System Overview

Architecture Scope

The **Village Education Resource Center (VERC)** web platform is an enterprise-grade digital ecosystem built to serve beneficiaries, donors, humanitarian partners (e.g. UNICEF, Water.org), and administrative stakeholders. The platform delivers high-speed public awareness, dynamic programmatic discovery (WaSH, Non-Formal Education, Health, Microfinance), an AI-assisted conversational interface powered by Google Gemini, and a centralized administrative control center with real-time optimistic state management.

2. End-to-End System Technical Architecture

Layered Stack

VERC PLATFORM MULTI-TIER ARCHITECTURE



Client/Edge



Frontend (Next.js)



Backend (NestJS)



Data/Persistence

1. CLIENT TIER

Omni-Channel

Public Web Visitors

Desktop & Mobile Responsive Browsers (Chrome, Safari, Edge, Firefox)

Admin Portal Users

Role-authenticated Content Managers & Program Officers

Donors & Partners

UN Agencies, International NGOs & Microfinance clients

2. EDGE & FRONTEND

Next.js 14 App Router

Server-Side Rendering (SSR)

Fast LCP / SEO rendering with Next.js 14 App Router

ContentContext Engine

Unified React Context holding 15+ schemas with LocalStorage cache

AI Chatbot Client (SSE)

EventSource streaming consumer for real-time natural language answers

UI / UX Styling

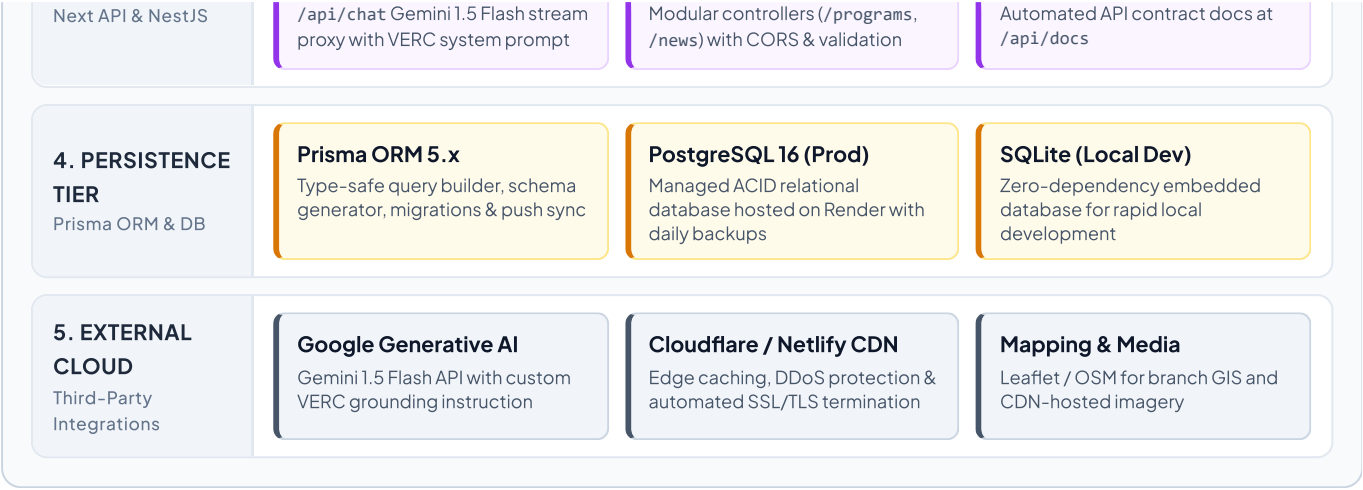
Tailwind CSS 3.4, Framer Motion animations & Lucide SVG Icons

3. API & MIDDLEWARE

Next.js Route Handlers

NestJS 10 REST API

OpenAPI / Swagger Docs

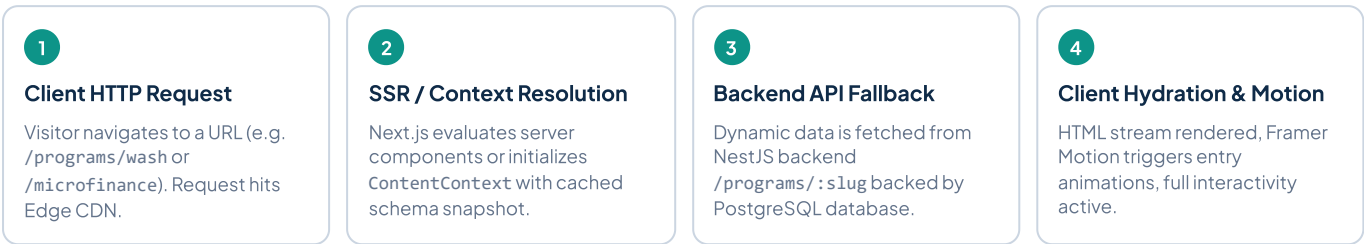


3. Core Operational Flowcharts

Execution Paths

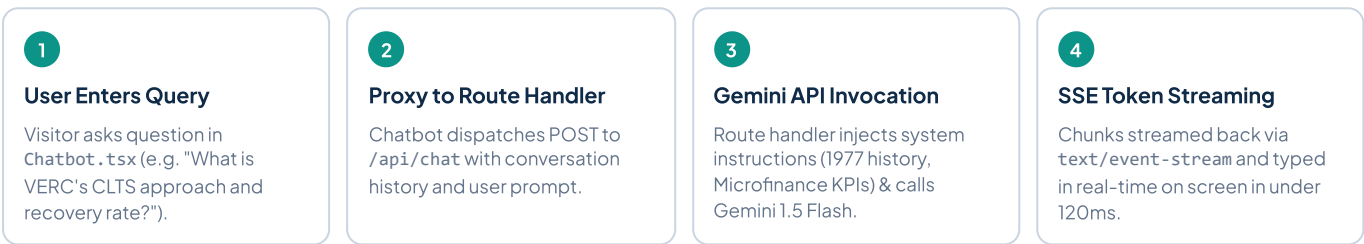
Flowchart A: Public Visitor Request & Content Rendering Flow

How visitor requests are routed, resolved, cached, and rendered with sub-second response times:



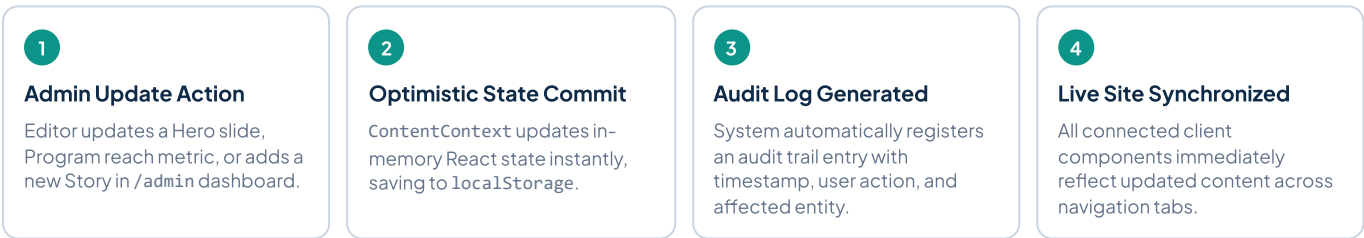
Flowchart B: AI Assistant Streaming Query Flow (Server-Sent Events)

The specialized low-latency flow connecting the visitor to the Google Gemini model with domain grounding:



Flowchart C: Admin Content Lifecycle & State Propagation

The administrative publishing workflow with immediate optimistic UI updates and persistent backend storage:



4. Database Schema & Data Models Specification

Entity Relationship

MODEL / ENTITY	PRIMARY FIELDS	DATA TYPES & CONSTRAINTS	OPERATIONAL USAGE
Program	id, title, slug, category, description, imageUrl, features	UUID (PK), slug (Unique), DateTime	Drives public programs directory (WaSH, Education, Health, MF) and detail pages.
SuccessStory	id, title, slug, content, location, imageUrl, date	UUID (PK), slug (Unique), DateTime	Impact showcase, field beneficiary transformation case studies, and donor reports.
News	id, title, summary, content, imageUrl, publishedAt, category	UUID (PK), DateTime (Indexed)	Organizational press releases, global visits (e.g. Orlando Bloom), workshop alerts.
Contact & Branch	id, officeName, address, phone, email, lat, lng	UUID (PK), Float (GPS lat/lng)	Branch locator mapping, public inquiry dispatching, and headquarters coordination.

MODEL / ENTITY	PRIMARY FIELDS	DATA TYPES & CONSTRAINTS	OPERATIONAL USAGE
MicrofinanceProduct	id, title, category, loanLimit, interestRate, tenure, eligibility	ContentContext Schema	Jagoron, Agrosor, Buniad, and Water Credit financial products calculation.

5. Operational Infrastructure & Deployment Blueprint

DevOps & Cloud

Production Cloud Hosting (Render + Netlify)

- Frontend Web Service:** Next.js 14 deployed on high-availability edge nodes with automatic cache invalidation on Git push.
- Backend Web Service:** Containerized NestJS 10 microservice running Node 20 runtime on Render with automated health checks.
- Database Engine:** Managed PostgreSQL 16 instance with connection pooling (DATABASE_URL) and SSL encryption.
- Declarative IaC:** Fully defined via render.yaml Blueprint for one-click environment provisioning.

Local Developer Environment

- 1-Click Scripts:** start-dev.bat / start-dev.sh to launch Frontend (:3000) and Backend (:3001) simultaneously.
- Local Persistence:** Embedded SQLite engine with instant migrations (prisma db push) and automated seed data.
- Container Option:** docker-compose.yml providing isolated PostgreSQL 16 with persistent volume verc_postgres_data.
- API Exploration:** Interactive Swagger UI at <http://localhost:3001/api/docs>.

Flowchart D: CI/CD Deployment & Database Migration Pipeline

1

Git Commit & Push

Developer pushes changes to main branch on GitHub repository.

2

Webhook Trigger

Render & Netlify webhooks detect push and initialize parallel container builds.

3

Build & Prisma Push

Executes npx prisma generate, compiles TypeScript, and synchronizes DB schema.

4

Zero-Downtime Swap

Health checks pass on new container; traffic shifts gracefully with zero downtime.

6. Security, Governance & Disaster Recovery

Enterprise Compliance

Security & Access Controls

- CORS Hardening:** Backend API restricts origin access to verified frontend domains.
- Secret Isolation:** GEMINI_API_KEY and database credentials stored exclusively in server environment variables, never leaked to client bundles.
- Input Sanitization:** Form inputs and API payloads are validated to prevent SQL injection and XSS.
- Audit Logging:** All administrative modifications (content edits, slide reorders) generate timestamped audit records.

Disaster Recovery & Reliability

- Automated Database Snapshots:** Managed daily snapshots with point-in-time recovery (PITR).
- Decoupled Resilience:** If backend API is temporarily unreachable, frontend utilizes client cached data for uninterrupted browsing.
- Graceful Degradation:** AI Assistant detects missing API keys or timeout and serves structured static help links.
- Data Portability:** Complete JSON export / import available inside Admin portal for instant state backup.

7. Verification & Sign-Off Checklist

Operational Readiness

VERIFICATION AREA	TARGET STANDARD	VALIDATION STATUS	TEST METHOD
Frontend Server	Next.js 14 App Router listening on :3000	VERIFIED (200 OK)	Automated HTTP check / SSR render test
Backend REST API	NestJS 10 endpoints (/programs, /news) on :3001	VERIFIED (200 OK)	PowerShell REST invoke / JSON validation
Swagger Documentation	OpenAPI interface live at /api/docs	VERIFIED (200 OK)	Interactive UI schema verification
Database & Seeds	Prisma ORM schema synchronized & seeded	VERIFIED (Active)	Prisma Client query inspection

VERIFICATION AREA	TARGET STANDARD	VALIDATION STATUS	TEST METHOD
AI Chatbot Gateway	Gemini 1.5 Flash SSE streaming proxy	CONFIGURED (/api/chat)	Route handler inspection & prompt grounding